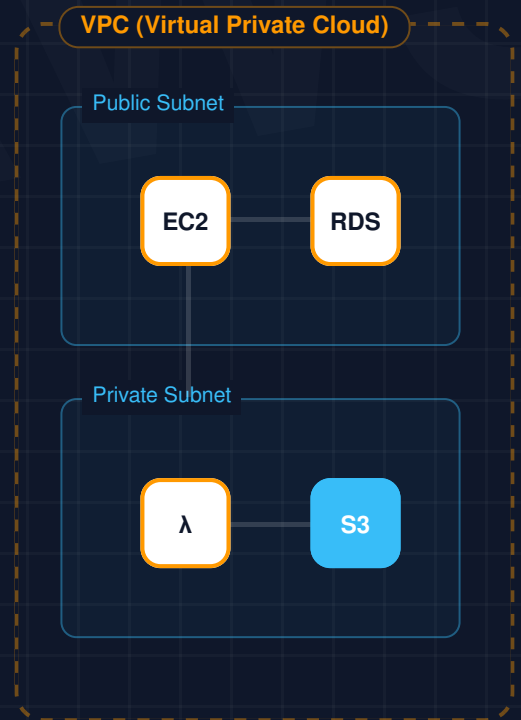


2026 EDITION

AWS INTERVIEW



Crack Cloud, DevOps & Solutions
Architect Interviews

- ✓ Most Asked AWS Questions
- ✓ Interviewer Approved Answers
- ✓ Freshers to Experienced Level
- ✓ AWS Fundamentals to Advanced Concepts

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Introduction to AWS in 2026

Amazon Web Services (AWS) remains the undisputed leader in cloud computing. In 2026, the demand for cloud professionals has evolved from simple infrastructure provisioning to architecting complex, resilient, and cost-optimized distributed systems. This guide, curated by **AlgoTutor**, is designed to give you the ultimate edge in your technical interviews.

Top AWS Career Paths

Solutions Architect

Designs high-availability architectures and translates business needs into AWS infrastructure.

DevOps Engineer

Automates CI/CD pipelines, manages infrastructure as code (IaC), and optimizes deployments.

Cloud Engineer

Focuses on the day-to-day operations, maintenance, and security of AWS cloud environments.

Interview Preparation Tips

- **Know the "Why":** Interviewers care less about standard definitions and more about *why* you would choose one service over another.
- **Master Cost Optimization:** In 2026, companies are heavily focused on FinOps. Mentioning cost impacts (e.g., NAT Gateway vs. VPC Endpoints) will score you major points.
- **Think Well-Architected:** Frame your answers around the 6 Pillars of the AWS Well-Architected Framework: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability.

Top 20 AWS Interview Questions

Q1 What is the key difference between Amazon EC2 and AWS Lambda?

Interviewer's Intent: Evaluates your understanding of IaaS (Infrastructure as a Service) versus FaaS (Function as a Service) / Serverless paradigms.

Answer: Amazon EC2 provides resizable virtual servers where you must manage the OS, scaling, patching, and provisioning. AWS Lambda is a serverless compute service that automatically runs code in response to events, managing all underlying compute resources for you.

Real-World Example: Hosting a legacy monolithic application requires an EC2 instance. Triggering a lightweight script to compress an image right after it's uploaded to S3 is best suited for Lambda.

Pro Tip: Always mention the pricing difference: EC2 bills per second/hour while running, whereas Lambda bills strictly per invocation and execution duration.

Key Takeaway: EC2 gives full control and persistence; Lambda offers event-driven, zero-maintenance execution.

Q2 Explain Amazon VPC and its core components.

Interviewer's Intent: Networking is the foundation of cloud security. Interviewers need to know you can isolate resources properly.

Answer: A Virtual Private Cloud (VPC) is a logically isolated virtual network in AWS. Its core components include Subnets (public/private resource segmentation), Route Tables (rules directing network traffic), Internet Gateways (allowing public access), NAT Gateways (allowing private resources outbound access), and Security Groups/NACLs (firewalls).

Real-World Example: A classic 3-tier web architecture places the Web servers in public subnets with an IGW, and the Database servers in private subnets with no direct internet access.

Pro Tip: Distinguish between Security Groups (stateful, instance-level) and NACLs (stateless, subnet-level).

Key Takeaway: VPC is your private data center in the cloud, defined by logical network boundaries.

Q3 What are the different Amazon S3 Storage Classes?

Interviewer's Intent: Tests your knowledge of cost-optimization and data lifecycle management.

Answer: S3 offers several tiers based on access frequency: S3 Standard (frequent access), S3 Intelligent-Tiering (auto-moves data based on access patterns), S3 Standard-IA (infrequent access but rapid retrieval), S3 One Zone-IA (lower cost, single AZ), S3 Glacier Instant/Flexible Retrieval (archival), and S3 Glacier Deep Archive (lowest cost, 12+ hours retrieval).

Real-World Example: A media company stores freshly uploaded videos in S3 Standard, then uses Lifecycle Policies to move them to Glacier Deep Archive after 1 year for compliance.

Pro Tip: Intelligent-Tiering is often the safest recommendation in interviews if data access patterns are unknown.

Key Takeaway: Match the S3 storage class to the specific durability, availability, and access requirements to optimize costs.

Q4 What is the difference between an IAM Role and an IAM Policy?

***Interviewer's Intent:** Identity and Access Management is highly scrutinized in interviews to ensure you follow security best practices.*

Answer: An IAM Policy is a JSON document that defines specific permissions (Allow/Deny to specific resources). An IAM Role is an identity that you can attach policies to, but unlike an IAM User, a Role does not have long-term credentials (passwords/access keys). It is meant to be assumed temporarily.

Real-World Example: You attach an IAM Role to an EC2 instance, allowing the application running on it to read from an S3 bucket without hardcoding AWS access keys in the source code.

Pro Tip: Mention the 'Principle of Least Privilege'—never grant full Admin access unless absolutely necessary.

Key Takeaway: Policies define **what** can be done; Roles define **who/what** temporarily assumes those permissions.

Q5 When would you choose Amazon RDS over Amazon DynamoDB?

Interviewer's Intent: *Assesses your database design architecture and ability to choose the right tool for the job.*

Answer: Amazon RDS is a managed relational database service ideal for structured data, complex multi-row transactions, and heavy SQL joins. DynamoDB is a fully managed NoSQL database designed for key-value document data, offering single-digit millisecond latency at any scale but lacking complex relational join capabilities.

Real-World Example: Use RDS for a financial banking ledger requiring ACID compliance and complex queries. Use DynamoDB for an online gaming leaderboard requiring massive, predictable read/write speeds.

Pro Tip: Mention 'Schema flexibility'. DynamoDB allows schema-less item creation, whereas RDS requires rigid schemas.

Key Takeaway: RDS is for relational SQL workloads; DynamoDB is for highly scalable NoSQL workloads.

Q6 How does Auto Scaling work in conjunction with Elastic Load Balancing (ELB)?

Interviewer's Intent: High availability and elasticity are core cloud concepts.

Answer: An Auto Scaling Group (ASG) automatically adds or removes EC2 instances based on demand (e.g., CPU utilization). An ELB automatically distributes incoming application traffic across those instances. When the ASG launches a new instance, it registers it with the ELB so it can immediately start receiving traffic.

Real-World Example: During a Black Friday sale, ASG scales out web servers to handle traffic spikes. ELB detects the new healthy instances and distributes the incoming shopper traffic among them seamlessly.

Pro Tip: Highlight health checks: ELB continually monitors instance health. If an instance fails, ELB stops sending traffic to it, and ASG terminates and replaces it.

Key Takeaway: ASG provides the computing power dynamically; ELB manages the traffic distribution.

Q7 Explain Amazon Route 53 and its routing policies.

Interviewer's Intent: Tests knowledge of DNS management and global traffic routing.

Answer: Route 53 is AWS's highly available DNS web service. It connects user requests to infrastructure. Key routing policies include Simple (standard DNS), Weighted (distribute traffic by percentage), Latency (route to region with lowest delay for user), Failover (Active/Passive disaster recovery), and Geolocation (route based on user's geographic location).

Real-World Example: Using Geolocation routing, European users are directed to the eu-central-1 region to comply with GDPR, while US users are directed to us-east-1.

Pro Tip: Route 53 is one of the few AWS services that guarantees 100% SLA for availability.

Key Takeaway: Route 53 is not just DNS; it's a global traffic management and failover orchestration tool.

Q8 What is Amazon CloudFront and how does it utilize Edge Locations?

Interviewer's Intent: Performance optimization and latency reduction are crucial for modern applications.

Answer: CloudFront is a global Content Delivery Network (CDN) service. It caches static and dynamic web content at Edge Locations—data centers spread across the world. When a user requests content, it is served from the nearest Edge Location, dramatically reducing latency and offloading traffic from the origin server.

Real-World Example: A streaming platform uses CloudFront to cache movie thumbnail images globally, ensuring users in Australia load the UI instantly without waiting for the primary S3 bucket in Virginia to respond.

Pro Tip: CloudFront isn't just for S3. It can be used with EC2, ELB, API Gateway, and even on-premise origins.

Key Takeaway: CloudFront improves global application performance by caching content close to the end-user.

Q9 What is the difference between a Public Subnet and a Private Subnet in a VPC?

Interviewer's Intent: This is a fundamental security and network architecture concept.

Answer: A Public Subnet has a route in its route table pointing to an Internet Gateway (IGW), allowing resources inside it to directly communicate with the public internet. A Private Subnet does not have a route to an IGW. To give resources in a Private Subnet internet access (e.g., for software updates), you must route their traffic through a NAT Gateway located in a Public Subnet.

Real-World Example: Web servers (which need to receive internet traffic) are placed in Public Subnets. Database servers (which hold sensitive data) are placed strictly in Private Subnets.

Pro Tip: Security Groups and NACLs provide defense in depth, but subnet routing is the architectural foundation of exposure.

Key Takeaway: Public subnets have direct IGW access; Private subnets require a NAT Gateway for outbound-only access.

Q10 Compare AWS CloudTrail and Amazon CloudWatch.

Interviewer's Intent: Monitoring, logging, and auditing are standard compliance requirements.

Answer: CloudWatch is a performance monitoring and management service (tracks metrics like CPU, Memory, Disk read/writes, and application logs). CloudTrail is an auditing and compliance service that records API calls made within your AWS account (tracks 'who did what, when, and from where').

Real-World Example: Use CloudWatch to trigger an alarm if an EC2 instance exceeds 80% CPU utilization. Use CloudTrail to investigate who accidentally deleted an important S3 bucket at 3:00 AM.

Pro Tip: Mnemonic: CloudWatch 'watches' performance. CloudTrail tracks the 'trail' of user actions.

Key Takeaway: CloudWatch = Metrics and Logging. CloudTrail = Auditing and API tracking.

Q11 When would you use Amazon SQS vs Amazon SNS?

Interviewer's Intent: *Microservices architectures rely heavily on decoupled communication.*

Answer: SQS (Simple Queue Service) is a message queue based on a pull/poll model. Consumers retrieve messages from the queue and process them at their own pace. SNS (Simple Notification Service) is a pub/sub messaging service based on a push model. It pushes messages to multiple subscribers simultaneously.

Real-World Example: Use SNS to broadcast an 'Order Placed' event. Use SQS to queue the backend jobs (like generating an invoice) so the processing servers can pull them without getting overwhelmed during a traffic spike.

Pro Tip: They are often used together: an SNS topic can push a single message to multiple SQS queues (Fanout pattern).

Key Takeaway: SQS is for reliable, asynchronous 1-to-1 processing (Pull). SNS is for 1-to-many broadcasting (Push).

Q12 How do AWS WAF and AWS Shield differ in protecting web applications?

***Interviewer's Intent:** Security engineers and architects must know how to mitigate common internet threats.*

Answer: AWS WAF (Web Application Firewall) protects against application-layer (Layer 7) attacks like SQL injection and Cross-Site Scripting (XSS) by allowing you to define custom security rules. AWS Shield is a managed DDoS protection service that protects against infrastructure-layer (Layer 3 & 4) volumetric attacks like UDP floods.

Real-World Example: If attackers are trying to brute-force a login endpoint using malicious SQL payloads, WAF blocks them. If a botnet tries to overwhelm your servers with raw junk traffic, Shield absorbs it.

Pro Tip: AWS Shield Standard is free and enabled by default for all AWS customers.

Key Takeaway: WAF blocks malicious application requests; Shield absorbs volumetric DDoS traffic.

Q13 What is an AWS Lambda 'Cold Start' and how can you mitigate it?

Interviewer's Intent: Tests practical experience with Serverless architectures and latency optimization.

Answer: A cold start occurs when a Lambda function is invoked after being idle. AWS must provision the execution environment, download the code, and initialize the runtime, causing a noticeable delay (latency).

Real-World Example: A user hitting an API endpoint connected to a dormant Lambda function might experience a 2-second delay, whereas subsequent requests (warm starts) execute in 50ms.

Pro Tip: To mitigate cold starts, you can use 'Provisioned Concurrency', which keeps a specified number of execution environments initialized and ready to respond immediately.

Key Takeaway: Cold starts impact latency; manage them using Provisioned Concurrency or by optimizing package size.

Q14 Describe the architecture for a highly available web application on AWS.

***Interviewer's Intent:** This is a classic 'whiteboard' interview question assessing overall system design.*

Answer: A highly available architecture leverages multiple Availability Zones (AZs). It starts with Route 53 routing traffic to an Application Load Balancer (ALB). The ALB distributes traffic across an Auto Scaling Group of EC2 instances spanning at least two AZs. The data tier uses Amazon RDS with a Multi-AZ deployment for synchronous replication and automatic failover.

Real-World Example: If a power outage takes down AZ-A, the ALB immediately stops routing traffic to instances in AZ-A. RDS automatically fails over to the standby database in AZ-B, keeping the app online.

Pro Tip: Don't forget to mention decoupling state. Store session data in ElastiCache or DynamoDB, and user uploads in S3.

Key Takeaway: High Availability means 'No Single Point of Failure'—use Multi-AZ and decouple application state.

Q15 What are the key differences between EBS, EFS, and S3?

Interviewer's Intent: AWS has many storage options. Selecting the wrong one leads to performance bottlenecks or cost overruns.

Answer: EBS (Elastic Block Store) is block storage attached to a single EC2 instance (like a local hard drive). EFS (Elastic File System) is a managed network file system that can be mounted by multiple EC2 instances simultaneously. S3 (Simple Storage Service) is an object storage service accessed via HTTP/API, not mounted directly as a filesystem.

Real-World Example: Use EBS for an OS boot drive or a database file. Use EFS for a shared content management system (CMS) directory used by multiple web servers. Use S3 for storing user-uploaded photos.

Pro Tip: EBS is highly performant but locked to a specific AZ. EFS is regional and shared.

Key Takeaway: EBS = Block (Single instance). EFS = File (Shared network). S3 = Object (API access).

Q16 What is AWS KMS and how does envelope encryption work?

***Interviewer's Intent:** Data encryption at rest is mandatory for enterprise workloads.*

Answer: AWS Key Management Service (KMS) lets you create and manage cryptographic keys. Because encrypting large amounts of data with a master key over the network is slow, AWS uses 'Envelope Encryption'. KMS generates a Data Key. The plaintext data is encrypted using this Data Key locally, and the Data Key itself is encrypted by the KMS Master Key.

Real-World Example: When an object is uploaded to an encrypted S3 bucket, S3 requests a Data Key from KMS, encrypts the object, and stores the encrypted object alongside the encrypted Data Key.

Pro Tip: Mention that KMS integrates natively with almost all AWS services (EBS, S3, RDS) requiring just a simple checkbox to enable.

Key Takeaway: KMS secures keys; Envelope Encryption optimizes the performance of encrypting large datasets.

Q17 How does AWS Fargate differ from Amazon EC2 when running containers?

Interviewer's Intent: Containerization (Docker/Kubernetes) is standard. Interviewers want to know how you run them on AWS.

Answer: When using ECS or EKS with EC2, you must provision, patch, and scale the underlying virtual machines that host your containers. AWS Fargate is a serverless compute engine for containers. It eliminates the need to manage EC2 instances; you simply specify the CPU and memory requirements, and AWS runs the container for you.

Real-World Example: A startup with no dedicated DevOps team chooses Fargate to run their Docker microservices so they don't have to worry about OS-level security patches on cluster nodes.

Pro Tip: Fargate shifts the shared responsibility model, pushing more of the infrastructure security burden to AWS.

Key Takeaway: EC2 = Manage the servers running containers. Fargate = Serverless container execution.

Q18 Explain the difference between On-Demand, Reserved, and Spot Instances.

Interviewer's Intent: FinOps is critical. You must know how to save money on compute.

Answer: On-Demand instances are billed by the second with no commitment. Reserved Instances (RIs) / Savings Plans require a 1 or 3-year commitment in exchange for massive discounts (up to 72%). Spot Instances let you bid on unused AWS capacity for steep discounts (up to 90%), but AWS can interrupt/terminate them with a 2-minute warning.

Real-World Example: Use On-Demand for unpredictable new workloads. Use RIs for a steady-state production database. Use Spot Instances for batch processing, image rendering, or fault-tolerant background jobs.

Pro Tip: Combine them in an Auto Scaling Group: keep baseline capacity as Reserved, and scale out during spikes using Spot.

Key Takeaway: On-Demand = Flexibility. Reserved = Baseline Cost Savings. Spot = Cheap but interruptible.

Q19 What are the best practices for securing an Amazon S3 bucket?

Interviewer's Intent: S3 data leaks are a common source of headline-making security breaches.

Answer: First, enable 'Block Public Access' at the account or bucket level. Use IAM policies and Bucket Policies to enforce least privilege. Enable default encryption at rest (SSE-S3 or SSE-KMS). Turn on Object Versioning and MFA Delete to prevent accidental or malicious deletion. Finally, use Macie or Access Analyzer to monitor for sensitive data exposure.

Real-World Example: A company suffered a data breach because a developer accidentally made a bucket public. 'Block Public Access' provides a hard guardrail against this.

Pro Tip: Never use S3 ACLs (Access Control Lists) in modern AWS architectures; use IAM and Bucket Policies instead.

Key Takeaway: Assume S3 buckets are private by default; aggressively lock them down and encrypt data at rest.

Q20 What is AWS CloudFormation and why is Infrastructure as Code (IaC) important?

Interviewer's Intent: Manual configuration in the console is frowned upon in enterprise environments. IaC is standard.

Answer: AWS CloudFormation allows you to model, provision, and manage AWS infrastructure using code (YAML or JSON templates). IaC is important because it makes infrastructure deployments repeatable, version-controlled (via Git), self-documenting, and free from human error.

Real-World Example: Instead of clicking through the AWS console to set up a VPC, subnets, and an EC2 instance for a QA environment, a developer runs a CloudFormation template that deploys the exact identical stack in minutes.

Pro Tip: Mention CloudFormation 'StackSets' which allow you to deploy infrastructure across multiple AWS accounts and regions simultaneously.

Key Takeaway: CloudFormation treats your infrastructure exactly like application source code.

Top AWS Services Every Candidate Must Know

Modern AWS architectures heavily rely on a core set of services. Mastering these is non-negotiable for passing a 2026 technical interview.

Service Name	Category	Primary Use Case
Amazon EC2	Compute	Virtual servers in the cloud. Granular control over OS.
Amazon S3	Storage	Scalable object storage for backups, static files, and data lakes.
AWS Lambda	Compute	Serverless event-driven computing.
Amazon RDS	Database	Managed relational databases (MySQL, PostgreSQL, etc.).
Amazon VPC	Networking	Isolated virtual network infrastructure.
AWS IAM	Security	Identity and access management (Users, Roles, Policies).
Amazon CloudWatch	Management	Monitoring resources and applications.
Amazon Route 53	Networking	Scalable Domain Name System (DNS) and traffic routing.

AWS Interview Cheat Sheet

Review these quick summaries right before stepping into your interview.

EC2 (Elastic Compute Cloud)

IaaS. You manage OS, patches, and scaling.
Remember the pricing models: On-Demand, Reserved, Spot.

S3 (Simple Storage Service)

Object storage. Remember the classes: Standard, IA, Intelligent-Tiering, Glacier. Data is highly durable (11 9s).

Lambda

Serverless FaaS. Max execution time is 15 mins. Billed per millisecond. Watch out for cold starts.

IAM (Identity & Access Mgmt)

Global service. Use Roles for EC2/Services. Follow the Principle of Least Privilege. Never share root keys.

VPC (Virtual Private Cloud)

Private network. Public subnets have IGW.
Private subnets need NAT Gateway to access the internet.

RDS (Relational Database)

Managed SQL. Use Multi-AZ for high availability (disaster recovery) and Read Replicas for scaling read performance.

AWS Interview Success Framework

1. BEFORE THE INTERVIEW

- Review the AWS Well-Architected Framework.
- Build a small project (e.g., host a static site on S3, or a simple serverless API using API Gateway + Lambda).
- Understand the pricing fundamentals.

2. DURING THE TECHNICAL ROUND

- **Clarify requirements:** Ask about expected traffic, data volume, and budget constraints before designing a solution.
- **Whiteboarding:** Always draw the VPC boundary first, then Subnets, then drop in compute/database resources.
- **Admit what you don't know:** AWS has 200+ services. It's okay to say, "I haven't used that service directly, but I know it's used for X."

3. HR / BEHAVIORAL ROUND

- Use the STAR Method (Situation, Task, Action, Result).
- Focus on moments where you optimized cloud costs or improved deployment security.

Thank You for Reading

Keep Learning. Keep Building. Keep Growing.

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